

A Meticulous Introduction to Stochastic Calculus

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August 2017

(ii): As $\Omega_0 \in \mathcal{F}_0$ and X_t^r is a right limit of X_t we find that X_t^r is adapted to $(\mathcal{F}_{t+})$ but since our filtration is right-continuous by assumption the conclusion holds.

(iii): Let $t \geq 0$ and pick a sequence $(t_n) \subseteq (t, \infty)$ such that $t_n \downarrow t$. Then we know from (i), Theorem 2.41 and uniqueness of limits that $X_{t_n} \rightarrow X_t^r$ almost surely and in $\mathcal{L}^1$. This yields that $X_t^r \in \mathcal{L}^1(\mathbb{P})$ and

$$\mathbb{E}[X_t^r] = \lim_{n \rightarrow \infty} \mathbb{E}[X_{t_n}],$$

hence $\mathbb{E}[X_t^r] = \lim_{s \downarrow t} \mathbb{E}[X_s]$ as our sequence (t_n) was arbitrary.

To show the last part we let $0 \leq s \leq t$ be given and use that conditional expectation respects convergence in $\mathcal{L}^1(\mathbb{P})$

$$\mathbb{E}[X_t^r | \mathcal{F}_s] = \lim_{n \rightarrow \infty} \mathbb{E}[X_{t_n} | \mathcal{F}_s] \geq X_s \quad \mathbb{P}\text{-a.s.} \quad (2.6)$$

where we can replace $\geq$ with $=$ if (X_t) is a martingale. Letting $s = t$ the result follows.

(iv): Let (s_n) be a sequence such that $s_n \in (s, t)$ and $s_n \downarrow s$. Then by (2.6)

$$\mathbb{E}[X_t^r | \mathcal{F}_{s_n}] \geq X_{s_n}, \quad \mathbb{P}\text{-a.s.} \quad (\text{with } = \text{ if martingale}).$$

The right-hand side converges to X_s^r almost surely and the left-hand side converges almost surely to $\mathbb{E}[X_t^r | \mathcal{F}_s]$ by Corollary 2.34 and using the filtration is right-continuous. This concludes the proof. $\square$

We say that two processes $(X_t)_{t \geq 0}$ and $(Y_t)_{t \geq 0}$ are **modifications** if

$$\forall t \geq 0 \exists A_t \in \mathcal{F} : X_t(\omega) = Y_t(\omega) \quad \text{for all } \omega \in A_t^c.$$

2.47 • Theorem (Martingale Modification theorem).

Let $(X_t, \mathcal{F}_t)$ be a submartingale and assume that the filtration satisfies the natural conditions. If $t \mapsto \mathbb{E}[X_t]$ is right-continuous, then $(X_t)_{t \geq 0}$ has a càdlàg modification, namely $(X_t^r)_{t \geq 0}$. In particular, if (X_t) is a martingale then it has a càdlàg modification.

Proof. By Lemma 2.46 (X_t^r) is a $(\mathcal{F}_t)$ -submartingale (or just a martingale) which is càdlàg. Furthermore we have the inequality $X_t^r \geq X_t$ almost surely and

$$\mathbb{E}[X_t^r] = \lim_{s \downarrow t} \mathbb{E}[X_s] = \mathbb{E}[X_t].$$

It follows that $\mathbb{E}[X_t^r - X_t] = 0$ and as $X_t^r - X_t \geq 0$ we must have $X_t^r = X_t$ almost surely, concluding the proof. $\square$

2.8 • Optional Sampling

In this section we want to extend the Optional Sampling theorems from discrete time to continuous time. We start with a lemma.

2.48 • Lemma. Let $(X_t, \mathcal{F}_t)$ be a progressively measurable submartingale with almost surely right-continuous sample paths. Let $\sigma \leq \tau$ be bounded extended stopping times and (τ_n) the discrete approximation of τ . Then $X_\tau, X_\sigma \in \mathcal{L}^1(\mathbb{P})$, $\mathbb{E}[X_\sigma] \leq \mathbb{E}[X_\tau]$ and $X_{\tau_n} \rightarrow X_\tau$ and $X_{\sigma_n} \rightarrow X_\sigma$ in $\mathcal{L}^1$.

As you read the proof you should notice that $\mathbb{E}[X_\sigma] = \mathbb{E}[X_\tau]$ if $(X_t, \mathcal{F}_t)$ is a martingale.

Proof. Note first that as X is progressively measurable and τ is bounded X_τ is measurable. The same goes for X_σ .

By assumption $X_{\tau_n} \rightarrow X_\tau$ almost surely so it is by symmetry enough to show that $\{X_{\tau_n} \mid n \in \mathbb{N}\}$ is uniformly integrable and $\mathbb{E}[X_{\sigma_n}] \leq \mathbb{E}[X_{\tau_n}]$ for all $n \in \mathbb{N}$. Let $\tilde{\tau}_n = 2^n \tau_n$ and $\tilde{\sigma}_n = 2^n \sigma_n$ and note that by Lemma 1.10

$$\{\tilde{\tau}_n = k\} = \{\tau_n = \frac{k}{2^n}\} \in \mathcal{F}_{k/2^n} \quad \text{for all } k \in \mathbb{N}_0$$

and likewise for $\tilde{\sigma}_n$. If we define $Z_k = X_{k/2^n}$ and $\mathcal{H}_k = \mathcal{F}_{k/2^n}$ then $(Z_k, \mathcal{H}_k)$ is a submartingale and $\tilde{\tau}_n, \tilde{\sigma}_n$ are bounded stopping times w.r.t. $\mathcal{H}_k$. Using that

$$\tilde{\sigma}_n = \sum_{k=1}^{\infty} k \mathbb{1}_{\{\sigma \in [\frac{k-1}{2^n}, \frac{k}{2^n})\}} \leq \sum_{k=1}^{\infty} \tilde{\tau}_n \mathbb{1}_{\{\sigma \in [\frac{k-1}{2^n}, \frac{k}{2^n})\}} = \tilde{\tau}_n$$

for all $n \in \mathbb{N}$ we get by Discrete Optional Sampling I that $Z_{\tilde{\tau}_n}, Z_{\tilde{\sigma}_n} \in \mathcal{L}^1(\mathbb{P})$ and $\mathbb{E}[Z_{\tilde{\sigma}_n}] \leq \mathbb{E}[Z_{\tilde{\tau}_n}]$. As

$$X_{\tau_n} = X_{\frac{\tilde{\tau}_n}{2^n}} = Z_{\tilde{\tau}_n} \quad \text{and likewise} \quad X_{\sigma_n} = Z_{\tilde{\sigma}_n}$$

we have that $X_{\tau_n}, X_{\sigma_n} \in \mathcal{L}^1(\mathbb{P})$ and $\mathbb{E}[X_{\sigma_n}] \leq \mathbb{E}[X_{\tau_n}]$.

We now only need to show uniform integrability. Define $Y_n = X_{\tau_{-n}}$ and $\mathcal{G}_n = \mathcal{F}_{\tau_{-n}}$ for $n \leq 0$. We want to show that (Y_n) is a $(\mathcal{G}_n)$ -submartingale. We have already showed integrability and adaptedness follows from progressively measurability of (X_t) . Hence we only need to show the submartingale property. For $n \in \mathbb{N}_0$ we find by Lemma 1.10 that

$$\begin{aligned} \mathcal{H}_{\tilde{\tau}_n} &= \{A \in \mathcal{F} \mid A \cap \{\tilde{\tau}_n = k\} \in \mathcal{H}_k \text{ for all } k \in \mathbb{N}_0\} \\ &= \{A \in \mathcal{F} \mid A \cap \{\tau_n = \frac{k}{2^n}\} \in \mathcal{F}_{\frac{k}{2^n}} \text{ for all } k \in \mathbb{N}_0\} \\ &= \mathcal{F}_{\tau_n}. \end{aligned}$$

So for $n \leq 0$ we can use Discrete Optional Sampling I on $(Z_k, \mathcal{H}_k)$ with the bounded stopping times $\tilde{\tau}_{n+1} \leq \tilde{\tau}_n$

$$\mathbb{E}[Y_n \mid \mathcal{G}_{n-1}] = \mathbb{E}[X_{\tau_{-n}} \mid \mathcal{F}_{\tau_{-n+1}}] = \mathbb{E}[Z_{\tilde{\tau}_{-n}} \mid \mathcal{H}_{\tilde{\tau}_{-n+1}}] \geq Z_{\tilde{\tau}_{-n+1}} = X_{\tau_{-n+1}} = Y_{n-1},$$

showing that $(Y_n, \mathcal{G}_n)_{n \leq 0}$ is a negatively index submartingale. We also have for all $n \leq 0$ that

$$\mathbb{E}[Z_0] \leq \mathbb{E}[Z_{\tilde{\tau}_{-n}}] = \mathbb{E}[Y_n] \leq \mathbb{E}[Y_0]$$

which implies that $\sup_{n \leq 0} \mathbb{E}[Y_n] < \infty$. By Theorem 2.32

$$\{Y_n \mid n \leq 0\} = \{X_{\tau_n} \mid n \in \mathbb{N}\}$$

is uniformly integrable. □

2.49 · Lemma. Let $(X_t, \mathcal{F}_t)$ a progressively measurable process and τ an extended stopping time with $X_\tau \in \mathcal{L}^1(\mathbb{P})$. Assume that if ξ is an extended stopping time satisfying $\xi \leq \tau$ then

$$X_\xi \in \mathcal{L}^1(\mathbb{P}) \quad \text{and} \quad \mathbb{E}[X_\xi] \leq \mathbb{E}[X_\tau].$$

Then for all extended stopping times σ with $\sigma \leq \tau$ we have

$$X_\sigma \leq \mathbb{E}[X_\tau | \mathcal{F}_{\sigma+}] \quad \mathbb{P}\text{-a.s.}$$

If σ is a stopping time we can replace $\mathcal{F}_{\sigma+}$ with $\mathcal{F}_\sigma$.

Proof. Let σ be an extended stopping time such that $\sigma \leq \tau$ and let $A \in \mathcal{F}_{\sigma+} \subseteq \mathcal{F}_{\tau+}$ and define

$$\tilde{\sigma}_A = \sigma \mathbb{1}_A + \tau \mathbb{1}_{A^c}.$$

Then $\tilde{\sigma}_A$ is an extended stopping time as

$$\{\tilde{\sigma}_A < t\} = \{\sigma < t\} \cap A \cup \{\tau < t\} \cap A^c \in \mathcal{F}_t.$$

We also have that $\tilde{\sigma}_A \leq \tau$ so by assumption $X_{\tilde{\sigma}_A} \in \mathcal{L}^1(\mathbb{P})$ and $\mathbb{E}[X_{\tilde{\sigma}_A}] \leq \mathbb{E}[X_\tau]$. This gives us

$$\begin{aligned} \mathbb{E}[X_\sigma \mathbb{1}_A] &= \mathbb{E}[X_{\tilde{\sigma}_A}] - \mathbb{E}[X_\tau \mathbb{1}_{A^c}] \leq \mathbb{E}[X_\tau] - \mathbb{E}[X_\tau \mathbb{1}_{A^c}] \\ &= \mathbb{E}[X_\tau \mathbb{1}_A] = \mathbb{E}[\mathbb{E}[X_\tau | \mathcal{F}_{\sigma+}] \mathbb{1}_A]. \end{aligned}$$

As this holds for all $A \in \mathcal{F}_{\sigma+}$ and X_σ is $\mathcal{F}_{\sigma+}$ -measurable it follows that

$$\{X_\sigma > \mathbb{E}[X_\tau | \mathcal{F}_{\sigma+}]\} \in \mathcal{F}_{\sigma+}$$

is a null set with respect to the restricted measure $\mathbb{P}|_{\mathcal{F}_{\sigma+}} : \mathcal{F}_{\sigma+} \rightarrow [0, 1]$. In particular it is a null set with respect to $\mathbb{P}$ so $X_\sigma \leq \mathbb{E}[X_\tau | \mathcal{F}_{\sigma+}]$ $\mathbb{P}$ -almost surely.

If σ is a stopping time then X_σ is $\mathcal{F}_\sigma$ -measurable hence by the Tower Property and the fact that $\mathcal{F}_\sigma \subseteq \mathcal{F}_{\sigma+}$

$$X_\sigma \leq \mathbb{E}[\mathbb{E}[X_\tau | \mathcal{F}_{\sigma+}] | \mathcal{F}_\sigma] = \mathbb{E}[X_\tau | \mathcal{F}_\sigma] \quad \mathbb{P}\text{-a.s.} \quad \square$$

We can now show Optional Sampling I. The result is only stated for submartingales but the result can be generalized to supermartingales by multiplying with -1 and reversing the inequalities and then to martingales using that martingales are both sub- and supermartingales and replacing the inequalities with equalities.

2.50 · Theorem (Optional Sampling I).

Let $(X_t, \mathcal{F}_t)$ be a progressively measurable submartingale with almost surely right-continuous sample paths. Then for every pair of bounded and extended $(\mathcal{F}_t)$ -stopping times $\sigma \leq \tau$ we have $X_\tau, X_\sigma \in \mathcal{L}^1(\mathbb{P})$ and

$$\mathbb{E}[X_0] \leq \mathbb{E}[X_\sigma] \leq \mathbb{E}[X_\tau] \quad \text{and} \quad X_\sigma \leq \mathbb{E}[X_\tau | \mathcal{F}_{\sigma+}] \quad \mathbb{P}\text{-a.s.}$$

If σ is a stopping time we can replace $\mathcal{F}_{\sigma+}$ with $\mathcal{F}_\sigma$ in the last inequality. Furthermore $(X_{\tau \wedge t}, \mathcal{F}_t)_{t \geq 0}$ is a progressively measurable submartingale for all extended stopping times τ .

Proof. By Lemma 2.48 we find using the extended stopping times $0 \leq \sigma$ and $\sigma \leq \tau$ that $X_\tau, X_\sigma \in \mathcal{L}^1(\mathbb{P})$ and

$$\mathbb{E}[X_0] \leq \mathbb{E}[X_\sigma] \leq \mathbb{E}[X_\tau].$$

By Lemma 2.49 we have (note that we may replace $\sigma+$ with σ by the same lemma if it is a stopping time)

$$X_\sigma \leq \mathbb{E}[X_\tau | \mathcal{F}_{\sigma+}] \quad \mathbb{P}\text{-a.s.}$$

To see the last part we note that $\tau \wedge t \leq t$ are bounded extended stopping times. So by the above $X_{\tau \wedge t}$ is in $\mathcal{L}^1(\mathbb{P})$. Moreover $(X_{t \wedge \tau})$ is progressively measurable by Theorem 1.9.

Let $0 \leq s < t$. We see using the above with the stopping times $s \leq t \wedge (s \vee \tau)$ that

$$\begin{aligned} \mathbb{E}[X_{\tau \wedge t} | \mathcal{F}_s] &= \mathbb{E}[X_{\tau \wedge s} \mathbb{1}_{\{\tau < s\}} | \mathcal{F}_s] + \mathbb{E}[X_{(\tau \vee s) \wedge t} \mathbb{1}_{\{\tau \geq s\}} | \mathcal{F}_s] \\ &= X_{\tau \wedge s} \mathbb{1}_{\{\tau < s\}} + \mathbb{1}_{\{\tau \geq s\}} \mathbb{E}[\mathbb{E}[X_{(\tau \vee s) \wedge t} | \mathcal{F}_{s+}] | \mathcal{F}_s] \\ &\geq X_{\tau \wedge s} \mathbb{1}_{\{\tau < s\}} + \mathbb{1}_{\{\tau \geq s\}} \mathbb{E}[X_s | \mathcal{F}_s] = X_{\tau \wedge s} \end{aligned}$$

almost surely, which concludes the proof. $\square$

Like in the discrete case we have the following definition.

2.51 • Definition. If $(\mathcal{F}_t)$ is a filtration and (X_t) a progressively measurable process. Then an extended $(\mathcal{F}_t)$ -stopping time τ is said to be **optional** for (X_t) if $\{X_{\tau \wedge n} | n \in \mathbb{N}_0\}$ is uniformly integrable.

Note that if τ is optional for (X_t) then $X_\tau \in \mathcal{L}^1(\mathbb{P})$: Measurability follows from the fact that (X_t) is progressively measurable, and integrability follows from Fatou's Lemma

$$\mathbb{E}[|X_\tau|] \leq \mathbb{E}[\liminf_{n \rightarrow \infty} |X_{\tau \wedge n}|] \leq \sup_{n \in \mathbb{N}_0} \mathbb{E}[|X_{\tau \wedge n}|] < \infty.$$

2.52 • Lemma. Let $(X_t, \mathcal{F}_t)$ be a progressively submartingale with almost surely right-continuous paths.

- (i) If τ is an extended stopping time then $(X_{\tau \wedge n}, \mathcal{F}_{(\tau \wedge n)+})_{n \in \mathbb{N}_0}$ is a discrete time submartingale.
- (ii) If τ is an extended stopping time and optional for (X_t^+) , then $X_\tau \in \mathcal{L}^1(\mathbb{P})$ and

$$X_{\tau \wedge n} \rightarrow X_\tau \quad \mathbb{P}\text{-a.s.} \quad \text{and} \quad X_{\tau \wedge n} \leq \mathbb{E}[X_\tau | \mathcal{F}_{(\tau \wedge n)+}] \quad \mathbb{P}\text{-a.s.}$$

Proof. (i): By Optional Sampling I $X_{\tau \wedge n} \in \mathcal{L}^1(\mathbb{P})$ for all $n \in \mathbb{N}_0$ and by progressive measurability $X_{\tau \wedge n}$ is $\mathcal{F}_{\tau \wedge n+}$ -measurable for all $n \in \mathbb{N}_0$. For n in $\mathbb{N}$ we have by Optional Sampling I that

$$\mathbb{E}[X_{\tau \wedge n} | \mathcal{F}_{\tau \wedge (n-1)+}] \geq X_{\tau \wedge (n-1)}$$

which proves (i).

(ii): As $(X_{\tau \wedge n}, \mathcal{F}_{(\tau \wedge n)+})_{n \geq 0}$ is a discrete time submartingale it follows from the assumption and Lemma 2.21 that $\sup_{n \geq 0} \mathbb{E}[|X_{\tau \wedge n}|] < \infty$. From the discrete time Martingale Convergence theorem we have $X_{\tau \wedge n} \rightarrow X_\tau$ $\mathbb{P}$ -a.s. From Corollary 2.29 we find that $X_\tau \in \mathcal{L}^1(\mathbb{P})$ and

$$X_{\tau \wedge n} \leq \mathbb{E}[X_\tau | \mathcal{F}_{(\tau \wedge n)+}] \quad \mathbb{P}\text{-a.s.}$$

concluding the proof. $\square$

We can now prove Option Sampling II in the submartingale case. We can extend the result to supermartingales and martingales the usual way.

2.53 · Theorem (Optional Sampling II).

Let $(X_t, \mathcal{F}_t)$ be a progressively measurable submartingale with almost surely right-continuous paths and let τ be an extended stopping time which is optional for (X_t^+) . Furthermore let σ be an extended stopping time such that $\sigma \leq \tau$. Then $X_\tau, X_\sigma \in \mathcal{L}^1(\mathbb{P})$, σ is optional for (X_t) and

$$\mathbb{E}[X_0] \leq \mathbb{E}[X_\sigma] \leq \mathbb{E}[X_\tau] \quad \text{and} \quad X_\sigma \leq \mathbb{E}[X_\tau | \mathcal{F}_{\sigma+}] \quad \mathbb{P}\text{-a.s.}$$

If σ is a stopping time one can replace $\mathcal{F}_{\sigma+}$ with $\mathcal{F}_\sigma$ in the last inequality.

Proof. By Lemma 2.52 $X_\tau \in \mathcal{L}^1(\mathbb{P})$. As $\sigma \wedge n \leq \tau \wedge n$ for all $n \in \mathbb{N}_0$ we get by Optional Sampling I and another use of Lemma 2.52 that

$$X_{\sigma \wedge n} \leq \mathbb{E}[X_{\tau \wedge n} | \mathcal{F}_{(\sigma \wedge n)+}] \leq \mathbb{E}[\mathbb{E}[X_\tau | \mathcal{F}_{(\tau \wedge n)+}] | \mathcal{F}_{(\sigma \wedge n)+}] = \mathbb{E}[X_\tau | \mathcal{F}_{(\sigma \wedge n)+}] \quad \mathbb{P}\text{-a.s.}$$

In particular

$$0 \leq X_{\sigma \wedge n}^+ \leq (\mathbb{E}[X_\tau | \mathcal{F}_{(\sigma \wedge n)+}])^+ \leq \mathbb{E}[X_\tau^+ | \mathcal{F}_{(\sigma \wedge n)+}] \quad \mathbb{P}\text{-a.s.}$$

This yields that $(X_{\sigma \wedge n}^+)$ is dominated by something uniformly integrable hence σ is optional for (X_t) . Lemma 2.52 then gives that $X_\sigma \in \mathcal{L}^1(\mathbb{P})$ and $X_{\sigma \wedge n} \rightarrow X_\sigma$ a.s.

We remind the reader that if $Y_n = \mathbb{E}[X | \mathcal{G}_n]$ then $Y_n \rightarrow \mathbb{E}[X | \mathcal{G}_\infty]$ almost surely and in $\mathcal{L}^1(\mathbb{P})$ where $\mathcal{G}_\infty = \sigma(\bigcup_{n=0}^\infty \mathcal{G}_n)$, cf. Theorem 2.30. This yields that

$$\mathbb{E}[X_\sigma] = \mathbb{E}[\lim_{n \rightarrow \infty} X_{\sigma \wedge n}] \leq \mathbb{E}[\lim_{n \rightarrow \infty} \mathbb{E}[X_\tau | \mathcal{F}_{(\sigma \wedge n)+}]] = \mathbb{E}[X_\tau].$$

The last part follows from Lemma 2.49. $\square$

2.9 · Martingales and natural filtrations

2.54 · Lemma. Let $(\mathcal{F}_t)$ and $(\mathcal{G}_t)$ be filtrations. If $\mathcal{F}_t \subseteq \mathcal{G}_t$ and (X_t) is adapted to $(\mathcal{F}_t)$ and a (sub-,super-)martingale with respect to $(\mathcal{G}_t)$, then (X_t) is a (sub-,super-)martingale with respect to $(\mathcal{F}_t)$.

Proof. As (X_t) is integrable and adapted to $(\mathcal{F}_t)$ we only need to show the martingale property which we only show in the submartingale case. But this follows from the tower property: Let $0 \leq s < t$ then

$$\mathbb{E}[X_t | \mathcal{F}_s] = \mathbb{E}[\mathbb{E}[X_t | \mathcal{G}_s] | \mathcal{F}_s] \geq \mathbb{E}[X_s | \mathcal{F}_s] = X_s \quad \mathbb{P}\text{-a.s.} \quad \square$$

(i): Let $\gamma < \frac{1}{2}$ and let $K \subseteq \mathbb{R}_+$ be compact and $s, t \in K$. We have that

$$\begin{aligned} |f(t) - f(s)| &= |a(g(t) - g(s)) + b(t - s)| \leq |a||g(t) - g(s)| + |b||t - s| \\ &\leq |a|C_g|t - s|^\gamma + |t - s|^{1-\gamma}|t - s|^\gamma \\ &\leq (|a|C_g + C)|t - s|^\gamma, \end{aligned}$$

using that g is locally Hölder continuous with exponent γ and that $|t - s|^{1-\gamma}$ is locally bounded.

(ii): Let $\gamma \in (\frac{1}{2}, 1]$, $C \in \mathbb{N}$, $\varepsilon > 0$ and $s \in \mathbb{R}_+$. Pick $C' \in \mathbb{N}$ so big that

$$|a|C' - |b|\varepsilon > C,$$

and find $t \in \mathbb{R}_+$ such that $|t - s| < \varepsilon$ and $|g(t) - g(s)| > C'|t - s|^\gamma$. We then have that

$$\begin{aligned} |f(t) - f(s)| &= |a(g(t) - g(s)) + b(t - s)| \geq |a||g(t) - g(s)| - |b||t - s| \\ &> |a|C'|t - s|^\gamma - |b||t - s|^{1-\gamma}|t - s|^\gamma \\ &> (|a|C' - |b|\varepsilon)|t - s|^\gamma > C|t - s|^\gamma, \end{aligned}$$

concluding the proof. $\square$

3.24 · Lemma. Let (B_t) be an $(\mathcal{F}_t)$ -Brownian motion and $c > 0$. Define the process $Y_t = B_{t+c} - B_c$. Then

- (i) (Y_t) is an $(\mathcal{F}_{t+c})_{t \geq 0}$ -Brownian motion.
- (ii) (Y_t) and (B_t) have the same distribution, i.e. $(B_t) \sim (Y_t)$.

Proof. This is Exercise 3.3. $\square$

3.4 · Stopping a Brownian motion

In this section we show two important results related to the Brownian motion, namely Wald's identity and the Reflection Principle.

3.25 · Theorem (Wald's Identity).

Let (B_t) be a standard $(\mathcal{F}_t)$ -Brownian motion and τ an extended $(\mathcal{F}_t)$ -stopping time. If $\mathbb{E}[\tau] < \infty$ then $\mathbb{E}[B_\tau^2] < \infty$,

$$\mathbb{E}[B_\tau] = 0 \quad \text{and} \quad \mathbb{E}[B_\tau^2] = \mathbb{E}[\tau].$$

Proof. Since (B_t) is a martingale it follows by Doob's Inequality that

$$\mathbb{E}[B_{t \wedge \tau}^2] \leq \mathbb{E}\left[\sup_{s \leq t} B_s^2\right] \leq 4\mathbb{E}[B_t^2] = 4t < \infty.$$

By Optional Sampling and Exercise 3.4 $M_t = B_{t \wedge \tau}^2 - t \wedge \tau$ is a martingale with mean 0, so in particular

$$\mathbb{E}[M_t] = \mathbb{E}[0] \quad \text{and} \quad \mathbb{E}[B_{t \wedge \tau}^2] = \mathbb{E}[t \wedge \tau].$$

By the martingale property we have for $m \leq n$ that

$$\begin{aligned}\mathbb{E}[(B_{n \wedge \tau} - B_{m \wedge \tau})^2] &= \mathbb{E}[B_{n \wedge \tau}^2 + B_{m \wedge \tau}^2 - 2B_{m \wedge \tau}B_{n \wedge \tau}] \\ &= \mathbb{E}[B_{n \wedge \tau}^2 - B_{m \wedge \tau}^2] \\ &= \mathbb{E}[n \wedge \tau - m \wedge \tau] \rightarrow 0 \quad \text{for } n, m \rightarrow \infty,\end{aligned}$$

using Dominated convergence (with 2τ). As such $(B_{n \wedge \tau})_{n \in \mathbb{N}}$ is Cauchy in $\mathcal{L}^2(\mathbb{P})$, and since continuity yields that $B_{n \wedge \tau} \rightarrow B_\tau$ a.s. we get by uniqueness of limits that $B_{n \wedge \tau} \rightarrow B_\tau$ in $\mathcal{L}^2(\mathbb{P})$. We conclude that $B_\tau \in \mathcal{L}^2(\mathbb{P})$ and by $\mathcal{L}^1(\mathbb{P})$ -convergence

$$\mathbb{E}[B_\tau] = \lim_{n \rightarrow \infty} \mathbb{E}[B_{\tau \wedge n}] = 0.$$

By $\mathcal{L}^2(\mathbb{P})$ -convergence it follows that

$$\mathbb{E}[B_\tau^2] = \lim_{n \rightarrow \infty} \mathbb{E}[B_{\tau \wedge n}^2] = \lim_{n \rightarrow \infty} \mathbb{E}[\tau \wedge n] = \mathbb{E}[\tau],$$

concluding the proof. □

3.26 • Corollary. Let (B_t) be a standard Brownian motion and $a, b \geq 0$ and $c \in \mathbb{R} \setminus \{0\}$. Define the stopping times

$$\tau_{a,b} = \inf\{t \geq 0 \mid B_t \in (-a, b)^c\} \quad \text{and} \quad \tau_c = \inf\{t \geq 0 \mid B_t = c\}.$$

We have the following results.

(i) $\mathbb{E}[\tau_{a,b}] = ab$ and

$$\mathbb{P}(B_{\tau_{a,b}} = -a) = \frac{b}{a+b} \quad \text{and} \quad \mathbb{P}(B_{\tau_{a,b}} = b) = \frac{a}{a+b}.$$

(ii) The stopping time τ_c is almost surely finite but $\mathbb{E}[\tau_c] = \infty$.

(iii) The Laplace transform of τ_c is given by

$$\mathbb{E}[e^{-\zeta \tau_c}] = e^{-\sqrt{2\zeta}|b|} \quad (\zeta \geq 0).$$

(iv) $\limsup_{t \rightarrow \infty} B_t = \infty$ and $\liminf_{t \rightarrow \infty} B_t = -\infty$ almost surely.

(ii) states that while the Brownian motion is almost surely certain to hit any given point the expected arrival time to this point is infinite.

Proof. (i): We note that $n \wedge \tau_{a,b}$ is a bounded stopping time and $|B_{\tau_{a,b} \wedge n}| \leq a + b$ for all $n \in \mathbb{N}$ so we find by Wald's Identity (cf. Theorem 3.25)

$$\mathbb{E}[\tau_{a,b} \wedge n] = \mathbb{E}[B_{\tau_{a,b} \wedge n}^2] \leq (a+b)^2.$$

Monotone Convergence then yields that $\mathbb{E}[\tau_{a,b}] \leq (a+b)^2 < \infty$. Using that $B_{\tau_{a,b}} \in \{-a, b\}$ a.s. by continuity, it follows that

$$\mathbb{P}(B_{\tau_{a,b}} = b) + \mathbb{P}(B_{\tau_{a,b}} = -a) = 1.$$

and another use of Wald's Identity yields

$$0 = \mathbb{E}[B_{\tau_{a,b}}] = b\mathbb{P}(B_{\tau_{a,b}} = b) - a\mathbb{P}(B_{\tau_{a,b}} = -a).$$

Solving the two equations with respect to the probabilities yields that

$$\mathbb{P}(B_{\tau_{a,b}} = -a) = \frac{b}{a+b} \quad \text{and} \quad \mathbb{P}(B_{\tau_{a,b}} = b) = \frac{a}{b+a}.$$

Finally we have that

$$\begin{aligned} \mathbb{E}[\tau_{a,b}] &= \mathbb{E}[B_{\tau_{a,b}}^2] = a^2 \mathbb{P}(B_{\tau_{a,b}} = -a) + b^2 \mathbb{P}(B_{\tau_{a,b}} = b) \\ &= \frac{a^2 b + ab^2}{a+b} = \frac{ab(a+b)}{(a+b)} = ab. \end{aligned}$$

This concludes (i).

(ii): Suppose $c > 0$ then

$$\mathbb{P}(\tau_c < \infty) \geq \mathbb{P}(B_{\tau_{n,c}} = c) = \frac{n}{c+n} \rightarrow 1 \quad \text{as } n \rightarrow \infty.$$

Likewise, if $c < 0$ then

$$\mathbb{P}(\tau_c < \infty) \geq \mathbb{P}(B_{\tau_{-c,n}} = c) = \frac{n}{-c+n} \rightarrow 1 \quad \text{as } n \rightarrow \infty.$$

Now if $\mathbb{E}[\tau_c] < \infty$ then Wald's Identity implies that $0 = \mathbb{E}[B_{\tau_c}] = c$, which is a contradiction.

(iii): For any $\alpha \in \mathbb{R}_+$ the process $(M_t^\alpha)_{t \geq 0}$ given by

$$M_t^\alpha = \exp(\alpha B_t - \frac{1}{2} \alpha^2 t), \quad (t \geq 0)$$

is a martingale by Exercise 3.4. Using Optional Sampling together with the bounded stopping time $t \wedge \tau_c$ it follows that

$$1 = \mathbb{E}[M_0^\alpha] = \mathbb{E}[M_{t \wedge \tau_c}^\alpha] = \mathbb{E}[\exp(\alpha B_{t \wedge \tau_c} - \frac{1}{2} \alpha^2 (t \wedge \tau_c))] \quad \text{for any } t \geq 0.$$

By (ii) τ_c is finite $\mathbb{P}$ -a.s hence

$$\lim_{n \rightarrow \infty} \exp(\alpha B_{n \wedge \tau_c} - \frac{1}{2} \alpha^2 (n \wedge \tau_c)) = \exp(\alpha B_{\tau_c} - \frac{1}{2} \alpha^2 \tau_c) \quad \mathbb{P}\text{-a.s.}$$

Suppose now that $c > 0$, then $B_{\tau_c \wedge n} \leq c$ for any n in $\mathbb{N}$ so $\exp(\alpha B_{n \wedge \tau_c} - \frac{1}{2} \alpha^2 (\tau_c \wedge n)) \leq \exp(\alpha c)$ and Dominated Convergence says that

$$1 = \lim_{n \rightarrow \infty} \mathbb{E}[\exp(\alpha B_{n \wedge \tau_c} - \frac{1}{2} \alpha^2 (n \wedge \tau_c))] = \mathbb{E}[\exp(\alpha B_{\tau_c} - \frac{1}{2} \alpha^2 \tau_c)] = e^{\alpha c} \mathbb{E}[e^{-\frac{1}{2} \alpha^2 \tau_c}].$$

Setting $\alpha = \sqrt{2c}$ the result now follows in the case $c > 0$.

If instead $c < 0$, then we consider the process $M_t^{-\alpha}$ (which is still a martingale by Exercise 3.4) and do the same calculations where we instead use that $B_{\tau_c \wedge n} \geq c$ and therefore Dominated Convergence with the bound $\exp(-\alpha c)$ implies that

$$1 = \lim_{n \rightarrow \infty} \mathbb{E}[\exp(-\alpha B_{\tau_c \wedge n} - \frac{1}{2} \alpha^2 \tau_c)] = e^{-\alpha c} \mathbb{E}[e^{-\frac{1}{2} \alpha^2 \tau_c}].$$

Chapter 3. Brownian motion

(iv): The inclusions:

$$\left\{ \sup_{t \geq 0} B_t < \infty \right\} \subseteq \bigcup_{n=1}^{\infty} \{\tau_n = \infty\} \quad \text{and} \quad \left\{ \inf_{t \geq 0} B_t > -\infty \right\} \subseteq \bigcup_{n=1}^{\infty} \{\tau_{-n} = \infty\},$$

imply that $\sup_{t \geq 0} B_t = \infty$ and $\inf_{t \geq 0} B_t = -\infty$ almost surely by (ii). Now for any n in $\mathbb{N}$ we have that

$$\sup_{t \geq n} B_t = \sup_{t \geq 0} B_{t+n} = B_n + \sup_{t \geq 0} (B_{t+n} - B_n) = \infty \quad \mathbb{P}\text{-a.s.}$$

as $(B_{t+n} - B_n)$ is a standard Brownian motion. Hence $\limsup_{t \rightarrow \infty} B_t = \infty$ a.s. and

$$\liminf_{t \rightarrow \infty} B_t = -\limsup_{t \rightarrow \infty} (-B_t) = -\infty \quad \mathbb{P}\text{-a.s.}$$

using that $(-B_t)$ is a standard Brownian motion. $\square$

The next big thing we want to prove is the Reflection Principle, however we need a small lemma.

3.27 • Lemma. Let (B_t) be a standard $(\mathcal{F}_t)$ -Brownian motion, τ an extended stopping time and let $t \geq 0$. Then for all $\theta \in \mathbb{R}$

$$\mathbb{E}[e^{i\theta(B_t - B_{t \wedge \tau})} \mid \mathcal{F}_{(\tau \wedge t)+}] = \mathbb{E}[e^{i\theta(B_{t \wedge \tau} - B_t)} \mid \mathcal{F}_{(\tau \wedge t)+}] \quad \mathbb{P}\text{-a.s.}$$

In particular $B_t - B_{t \wedge \tau} \sim B_{t \wedge \tau} - B_t$.

Proof. Let $A \in \mathcal{F}_{(\tau \wedge t)+}$ and let (τ_n) be the discrete approximation of $\tau \wedge t$ given in Lemma 1.10. Let $n \in \mathbb{N}$ then using that

$$A \cap \{\tau_n = \frac{k}{2^n}\} \in \mathcal{F}_{k/2^n}$$

and $B_t - B_{k/2^n}$ is independent of $\mathcal{F}_{\frac{k}{2^n}}$ for $t \geq \frac{k}{2^n}$, it follows that

$$\begin{aligned} \mathbb{E}[e^{i\theta(B_t - B_{t \wedge \tau_n})} \mathbb{1}_A] &= \sum_{k: \frac{k}{2^n} \leq t} \mathbb{E}[e^{i\theta(B_t - B_{k/2^n})} \mathbb{1}_{A \cap \{\tau_n = \frac{k}{2^n}\}}] \\ &= \sum_{k: \frac{k}{2^n} \leq t} \mathbb{E}[e^{i\theta(B_t - B_{k/2^n})}] \mathbb{P}(A \cap \{\tau_n = \frac{k}{2^n}\}) \\ &= \sum_{k: \frac{k}{2^n} \leq t} \mathbb{E}[e^{i\theta(B_{k/2^n} - B_t)}] \mathbb{P}(A \cap \{\tau_n = \frac{k}{2^n}\}) \\ &= \sum_{k: \frac{k}{2^n} \leq t} \mathbb{E}[e^{i\theta(B_{k/2^n} - B_t)} \mathbb{1}_{A \cap \{\tau_n = \frac{k}{2^n}\}}] \\ &= \mathbb{E}[e^{i\theta(B_{t \wedge \tau_n} - B_t)} \mathbb{1}_A]. \end{aligned}$$

Hence by continuity and Dominated convergence

$$\begin{aligned} \mathbb{E}[e^{i\theta(B_t - B_{t \wedge \tau})} \mathbb{1}_A] &= \lim_{n \rightarrow \infty} \mathbb{E}[e^{i\theta(B_t - B_{t \wedge \tau_n})} \mathbb{1}_A] = \lim_{n \rightarrow \infty} \mathbb{E}[e^{i\theta(B_{t \wedge \tau_n} - B_t)} \mathbb{1}_A] \\ &= \mathbb{E}[e^{i\theta(B_{t \wedge \tau} - B_t)} \mathbb{1}_A], \end{aligned}$$

concluding the proof. $\square$

The Reflection Principle states that, if we take a process that follows the path of a Brownian motion up to a stopping time and then reflect the Brownian motion around the value at this point, we get a Brownian motion. This is easily visualised in Figure 3.3.

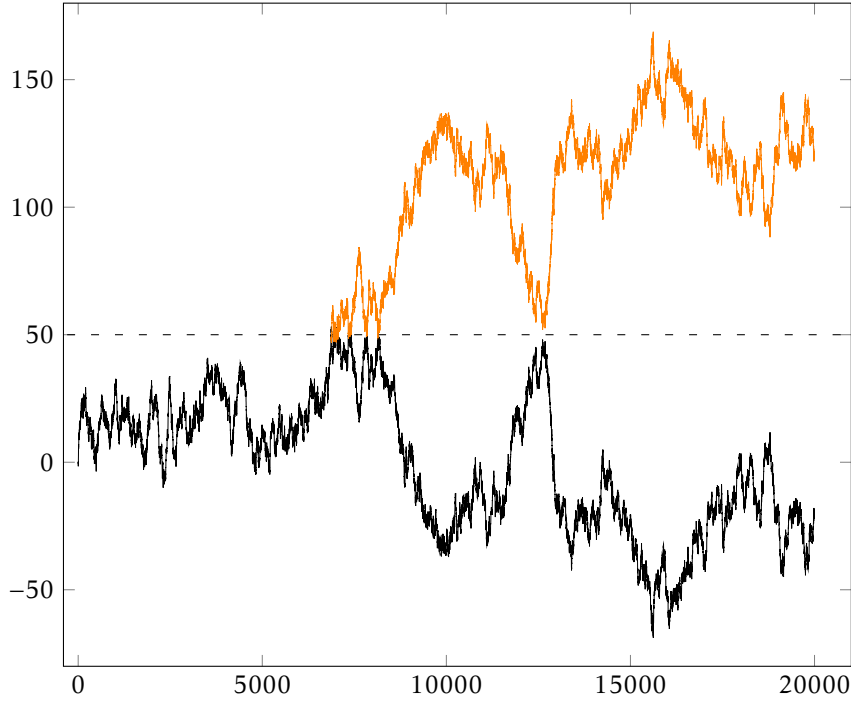


Figure 3.3: The Reflection Principle.

3.28 · Theorem (Reflection Principle).

Let (B_t) be a standard $(\mathcal{F}_t)$ -Brownian motion and τ an almost surely finite extended $(\mathcal{F}_t)$ -stopping time. Then the process given by

$$W_t = B_t \mathbb{1}_{\{\tau > t\}} + (2B_\tau - B_t) \mathbb{1}_{\{\tau \leq t\}}.$$

is a standard $(\mathcal{F}_t)$ -Brownian motion.

Proof. First we note that for $t \geq 0$ and $A \in \mathcal{B}(\mathbb{R})$ then

$$\{B_\tau \mathbb{1}_{\{\tau \leq t\}} \in A\} = \{B_\tau \mathbb{1}_{\{\tau < \infty\}} \in A\} \cap \{\tau \leq t\} \cup \{0 \in A\} \cap \{\tau > t\} \in \mathcal{F}_t$$

since $B_\tau \mathbb{1}_{\{\tau < \infty\}}$ is $\mathcal{F}_\tau$ -measurable by Theorem 1.9. It follows that (W_t) is adapted to $(\mathcal{F}_t)$. Since (B_t) is continuous and $W_\tau = B_\tau$ we conclude that (W_t) is continuous and also note that $W_0 = 0$ almost surely. We therefore only need to check that (W_t) has independent and normally distributed increments.

Let $s \leq t$ and $A \in \mathcal{F}_s$ then

$$\mathbb{E}[e^{i\theta(W_t - W_s)} \mathbb{1}_A] = \mathbb{E}[e^{i\theta(W_t - W_s)} \mathbb{1}_{\{\tau > s\} \cap A}] + \mathbb{E}[e^{i\theta(W_t - W_s)} \mathbb{1}_{\{\tau \leq s\} \cap A}].$$

The first term is

$$\begin{aligned} \mathbb{E}[e^{i\theta(W_t - W_s)} \mathbb{1}_{\{\tau > s\} \cap A}] &= \mathbb{E}[e^{i\theta(W_t - W_s)} \mathbb{1}_{\{\tau > s, \tau \leq t\} \cap A}] + \mathbb{E}[e^{i\theta(W_t - W_s)} \mathbb{1}_{\{\tau > t\} \cap A}] \\ &= \mathbb{E}[e^{i\theta(2B_\tau - B_t - B_s)} \mathbb{1}_{\{\tau > s, \tau \leq t\} \cap A}] + \mathbb{E}[e^{i\theta(B_t - B_s)} \mathbb{1}_{\{\tau > t\} \cap A}]. \end{aligned}$$

Chapter 3. Brownian motion

Let $v > 0$ then

$$\begin{aligned} A \cap \{\tau \in (s, t]\} \cap \{\tau \wedge t < v\} &= A \cap \{\tau \in (s, t]\} \cap \{\tau < v\} \\ &= \begin{cases} \emptyset \in \mathcal{F}_v, & \text{if } v \leq s, \\ A \cap \{\tau \in (s, t]\} \cap \{\tau < v\} \in \mathcal{F}_v, & \text{if } v > s. \end{cases} \end{aligned}$$

hence $A \cap \{\tau \in (s, t]\} \in \mathcal{F}_{(\tau \wedge t)+}$. Lemma 3.27 now that yields

$$\begin{aligned} \mathbb{E}[e^{i\theta(2B_\tau - B_t - B_s)} \mathbb{1}_{\{\tau \in (s, t]\} \cap A}] &= \mathbb{E}[e^{i\theta(2B_{\tau \wedge t} - B_t - B_{s \wedge \tau})} \mathbb{1}_{\{\tau \in (s, t]\} \cap A}] \\ &= \mathbb{E}[\mathbb{1}_{\{\tau \in (s, t]\} \cap A} e^{i\theta(B_{t \wedge \tau} - B_{s \wedge \tau})} \mathbb{E}[e^{i\theta(B_{\tau \wedge t} - B_t)} \mid \mathcal{F}_{(t \wedge \tau)+}]] \\ &= \mathbb{E}[\mathbb{1}_{\{\tau \in (s, t]\} \cap A} e^{i\theta(B_{t \wedge \tau} - B_{s \wedge \tau})} \mathbb{E}[e^{i\theta(B_t - B_{\tau \wedge t})} \mid \mathcal{F}_{(t \wedge \tau)+}]] \\ &= \mathbb{E}[e^{i\theta(B_t - B_s)} \mathbb{1}_{\{\tau \in (s, t]\} \cap A}]. \end{aligned}$$

So we have that

$$\mathbb{E}[e^{i\theta(W_t - W_s)} \mathbb{1}_{\{\tau > s\} \cap A}] = \mathbb{E}[e^{i\theta(B_t - B_s)} \mathbb{1}_{\{\tau \in (s, t]\} \cap A}] + \mathbb{E}[e^{i\theta(B_t - B_s)} \mathbb{1}_{\{\tau > t\} \cap A}].$$

Using symmetry and independence the second term is

$$\begin{aligned} \mathbb{E}[e^{i\theta(W_t - W_s)} \mathbb{1}_{\{\tau \leq s\} \cap A}] &= \mathbb{E}[e^{i\theta(2B_\tau - B_t - 2B_\tau + B_s)} \mathbb{1}_{\{\tau \leq s\} \cap A}] \\ &= \mathbb{E}[e^{i\theta(-B_t + B_s)} \mathbb{1}_{\{\tau \leq s\} \cap A}] \\ &= \mathbb{E}[e^{i\theta(B_t - B_s)} \mathbb{1}_{\{\tau \leq s\} \cap A}]. \end{aligned}$$

So it follows that

$$\mathbb{E}[e^{i\theta(W_t - W_s)} \mathbb{1}_A] = \mathbb{E}[e^{i\theta(B_t - B_s)} \mathbb{1}_A] = \mathbb{P}(A) \mathbb{E}[e^{i\theta B_{t-s}}].$$

Picking $A = \Omega$ yields that $W_t - W_s \sim N(0, t - s)$ and Exercise 3.2 says that $W_t - W_s$ is independent of $\mathcal{F}_s$. We conclude that (W_t) is a standard Brownian motion. $\square$

The next two results are related to the supremum of a Brownian motion and it turns out that the supremum has a nice distribution, which we now prove.

3.29 • Theorem. Let (B_t) be a Brownian motion and $u, c > 0$. Then

$$\mathbb{P}\left(\sup_{s \leq u} B_s \geq c\right) = 2\mathbb{P}(B_u \geq c) = \mathbb{P}(|B_u| \geq c).$$

Proof. Let $(\mathcal{F}_t^B)$ be the filtration generated by (B_t) and define the $(\mathcal{F}_t^B)$ -stopping time $\tau_c = \inf\{t \geq 0 \mid B_t = c\}$. By Corollary 3.26 we get that τ_c is an almost surely finite stopping time, furthermore we have by continuity

$$\left\{\sup_{t \leq s} B_t \geq c\right\} \cap \{B_0 = 0\} = \{\tau_c \leq s\} \cap \{B_0 = 0\}.$$

Define the process $(W_t)_{t \geq 0}$ by the expression

$$W_t = B_t \mathbb{1}_{\{\tau_c > t\}} + (2B_{\tau_c} - B_t) \mathbb{1}_{\{\tau_c \leq t\}},$$

and note that (W_t) is a standard $(\mathcal{F}_t^B)$ -Brownian motion by the Reflection Principle (Theorem 3.28). For all $s \geq 0$ we see that

$$\begin{aligned} \mathbb{P}\left(\sup_{t \leq s} B_t \geq c\right) &= \mathbb{P}\left(\sup_{t \leq s} B_t \geq c, B_s > c\right) + \mathbb{P}\left(\sup_{t \leq s} B_t \geq c, B_s \leq c\right) \\ &= \mathbb{P}(B_s \geq c) + \mathbb{P}\left(\sup_{t \leq s} B_t \geq c, B_s \leq c\right). \end{aligned}$$

If we can show that

$$\mathbb{P}\left(\sup_{t \leq s} B_t \geq c, B_s \leq c\right) = \mathbb{P}(W_s \geq c)$$

the proof will be done as the equality $\mathbb{P}(W_s \geq c) = \mathbb{P}(B_s \geq c)$ will yield

$$\mathbb{P}\left(\sup_{t \leq s} B_t \geq c\right) = \mathbb{P}(B_s \geq c) + \mathbb{P}(B_s \geq c) = 2\mathbb{P}(B_s \geq c).$$

Let $\omega \in \{\sup_{t \leq s} B_t \geq c, B_s \leq c\} \cap \{B_{\tau_c} = c\}$ then $\tau_c(\omega) \leq s$ and

$$W_s(\omega) = 2c - B_s(\omega) \geq 2c - c = c.$$

On the other hand if $\omega \in \{W_s \geq c\} \cap \{B_{\tau_c} = c\}$ we have that $\tau_c(\omega) \leq s$: Assume otherwise that $\tau_c(\omega) > s$ then

$$c \leq W_s(\omega) = B_s(\omega),$$

which is a contradiction since this implies that $\tau_c(\omega) \leq s$. Hence $\sup_{t \leq s} B_s(\omega) \geq c$ and

$$c \leq W_s(\omega) = 2c - B_s(\omega) \implies B_s(\omega) \leq c$$

so we conclude that

$$\left\{\sup_{t \leq s} B_t \geq c, B_s \leq c\right\} \cap \{B_{\tau_c} = c\} = \{W_s \geq c\} \cap \{B_{\tau_c} = c\}.$$

Since $\mathbb{P}(B_{\tau_c} = c) = 1$ the result follows. $\square$

The next proposition tells us that a Brownian motion will, with positive probability, stay in an ε -belt around 0. Later, in Chapter 10, we will show that a Brownian motion looks like any continuous function with positive probability.

3.30 · Proposition. Let (B_t) be an $(\mathcal{F}_t)$ -Brownian motion. Then for all $\varepsilon > 0$ and $t \geq 0$ it holds that

$$\mathbb{P}\left(\sup_{s \leq t} |B_s| < \varepsilon\right) > 0.$$

Proof. As $B_0 = 0$ a.s. and $t \mapsto B_t$ is continuous we find that

$$\mathbb{P}\left(\left\{\sup_{s \leq t} |B_s| \leq \varepsilon\right\} \cap \{B_0 = 0\}\right) \uparrow 1 \quad \text{for } t \rightarrow 0.$$

Hence there exists a $\delta > 0$ such that

$$\mathbb{P}\left(\sup_{s \leq \delta} |B_s| \leq \frac{\varepsilon}{2}\right) = 2\alpha > 0.$$

Since $(B_t) \sim (-B_t)$ we have that

$$\begin{aligned} \mathbb{P}\left(\sup_{s \leq \delta} |B_s| \leq \frac{\varepsilon}{2}\right) &= \mathbb{P}\left(\sup_{s \leq \delta} |B_s| \leq \frac{\varepsilon}{2}, B_\delta \leq 0\right) + \mathbb{P}\left(\sup_{s \leq \delta} |B_s| \leq \frac{\varepsilon}{2}, B_\delta > 0\right) \\ &= 2\mathbb{P}\left(\sup_{s \leq \delta} |B_s| \leq \frac{\varepsilon}{2}, B_\delta \leq 0\right) \\ &= 2\mathbb{P}\left(\sup_{s \leq \delta} |B_s| \leq \frac{\varepsilon}{2}, B_\delta > 0\right). \end{aligned}$$

Chapter 3. Brownian motion

This yields for all $x \in [-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}]$ that

$$\begin{aligned} \mathbb{P}\left(\sup_{s \leq \delta} |B_s + x| \leq \varepsilon, x + B_\delta \in [-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}]\right) &\geq \begin{cases} \mathbb{P}\left(\sup_{s \leq \delta} |B_s| \leq \frac{\varepsilon}{2}, B_\delta \leq 0\right), & \text{for } x \geq 0, \\ \mathbb{P}\left(\sup_{s \leq \delta} |B_s| \leq \frac{\varepsilon}{2}, B_\delta > 0\right), & \text{for } x < 0, \end{cases} \\ &= \frac{1}{2} \mathbb{P}\left(\sup_{s \leq \delta} |B_s| \leq \frac{\varepsilon}{2}\right). \end{aligned}$$

Assume for induction that $\mathbb{P}(\sup_{s \leq n\delta} |B_s| \leq \varepsilon, B_{n\delta} \in [-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}]) \geq \alpha^n$ for some $n \in \mathbb{N}$, where the case $n = 1$ follow from the above calculations with $x = 0$.

Using that $\sup_{s \leq n\delta} |B_s|$ is $\mathcal{F}_{n\delta}$ -measurable and $Y_t := B_{t+n\delta} - B_{n\delta}$ is an $(\mathcal{F}_{n\delta+t})$ -Brownian motion, in particular it is independent of $\mathcal{F}_{n\delta}$, it follows that

$$\begin{aligned} &\mathbb{P}\left(\sup_{s \leq (n+1)\delta} |B_s| \leq \varepsilon, B_{(n+1)\delta} \in [-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}]\right) \\ &= \mathbb{P}\left(\sup_{s \leq n\delta} |B_s| \leq \varepsilon, \sup_{s \in [n\delta, (n+1)\delta]} |B_s| \leq \varepsilon, B_{n\delta+\delta} \in [-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}]\right) \\ &= \mathbb{E}\left[\mathbb{1}_{\{\sup_{s \leq n\delta} |B_s| \leq \varepsilon\}} \mathbb{1}_{[0, \varepsilon]} \left(\sup_{s \leq \delta} |B_{s+n\delta}|\right) \mathbb{1}_{[-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}]}(B_{n\delta+\delta})\right] \\ &= \mathbb{E}\left[\mathbb{1}_{\{\sup_{s \leq n\delta} |B_s| \leq \varepsilon\}} \mathbb{1}_{[0, \varepsilon]} \left(\sup_{s \leq \delta} |B_{n\delta} + Y_s|\right) \mathbb{1}_{[-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}]}(B_{n\delta} + Y_\delta)\right] \\ &= \mathbb{E}\left[\mathbb{1}_{\{\sup_{s \leq n\delta} |B_s| \leq \varepsilon\}} \mathbb{P}\left(\sup_{s \leq \delta} |B_{n\delta} + Y_s| \leq \varepsilon, B_{n\delta} + Y_\delta \in [-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}] \mid \mathcal{F}_{n\delta}\right)\right] \\ &= \mathbb{E}[\mathbb{1}_{\{\sup_{s \leq n\delta} |B_s| \leq \varepsilon\}} F(B_{n\delta})], \end{aligned}$$

where $F : \mathbb{R} \rightarrow [0, 1]$ is the function given by

$$F(x) = \mathbb{P}\left(\sup_{s \leq \delta} |x + Y_s| \leq \varepsilon, x + Y_\delta \in [-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}]\right).$$

Since $(Y_t) \sim (B_t)$ it follows that

$$\inf_{x \in [-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}]} F(x) = \inf_{x \in [-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}]} \mathbb{P}\left(\sup_{s \leq \delta} |x + B_s| \leq \varepsilon, x + B_\delta \in [-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}]\right) \geq \alpha$$

So we get that

$$\begin{aligned} \mathbb{P}\left(\sup_{s \leq (n+1)\delta} |B_s| \leq \varepsilon\right) &\geq \mathbb{E}\left[\mathbb{1}_{\{\sup_{s \leq n\delta} |B_s| \leq \varepsilon\}} \mathbb{1}_{\{B_{n\delta} \in [-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}]\}} F(B_{n\delta})\right] \\ &\geq \alpha \mathbb{P}\left(\sup_{s \leq n\delta} |B_s| \leq \varepsilon, B_{n\delta} \in [-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}]\right) \geq \alpha^{n+1}, \end{aligned}$$

concluding the induction step. Letting $t \geq 0$ and $\varepsilon > 0$ be fixed we find an $n \in \mathbb{N}$ such that $n\delta > t$ and

$$\mathbb{P}\left(\sup_{s \leq n\delta} |B_s| \leq \frac{\varepsilon}{2}\right) > 0$$

which implies that

$$\mathbb{P}\left(\sup_{s \leq t} |B_s| < \varepsilon\right) \geq \mathbb{P}\left(\sup_{s \leq n\delta} |B_s| \leq \frac{\varepsilon}{2}\right) > 0,$$

concluding the proof. $\square$